

ML Driven Mortgage Prepayment and MBS Cash-Flow Analytics

1. Data, Sampling and Target Design

Business question - Can borrower-level machine-learning models improve our understanding and forecasting of mortgage prepayments and how do those changing prepayment expectations affects the timing of cash flows received by MBS investor?

Data Engineering approaches-

- Ingested Freddie Mac 2020 to 2023 quarterly origination and performance files and converted raw pipe delimited text into **parquet** for efficient analytical processing.
- Sampled at the loan level so that each selected mortgage retained its full observed monthly history.
- Selected 9,375 from each of 16 origination quarters, producing 150,000 unique loans and 6,418,477 loan-month records spanning Jan 2020 through March 2026. While random sampling selected `random_state = 42` so that this random sampling is repeatable and did not just take the first 9,375 loans by `loan_sequence_number`.
- Joined historical Freddie Mac Primary Mortgage Market Survey (PMMS) 30-year mortgage rates so each monthly observation and engineered refinance incentive as `current mortgage rates minus prevailing PMMS`.

Target: next-month voluntary prepayment

For each eligible active loan-month, `prepay_1m_final = 1` when the mortgage voluntarily prepays in the following month. The model estimates the probability of that binary event. Termination month rows are not treated as eligible prediction observations.

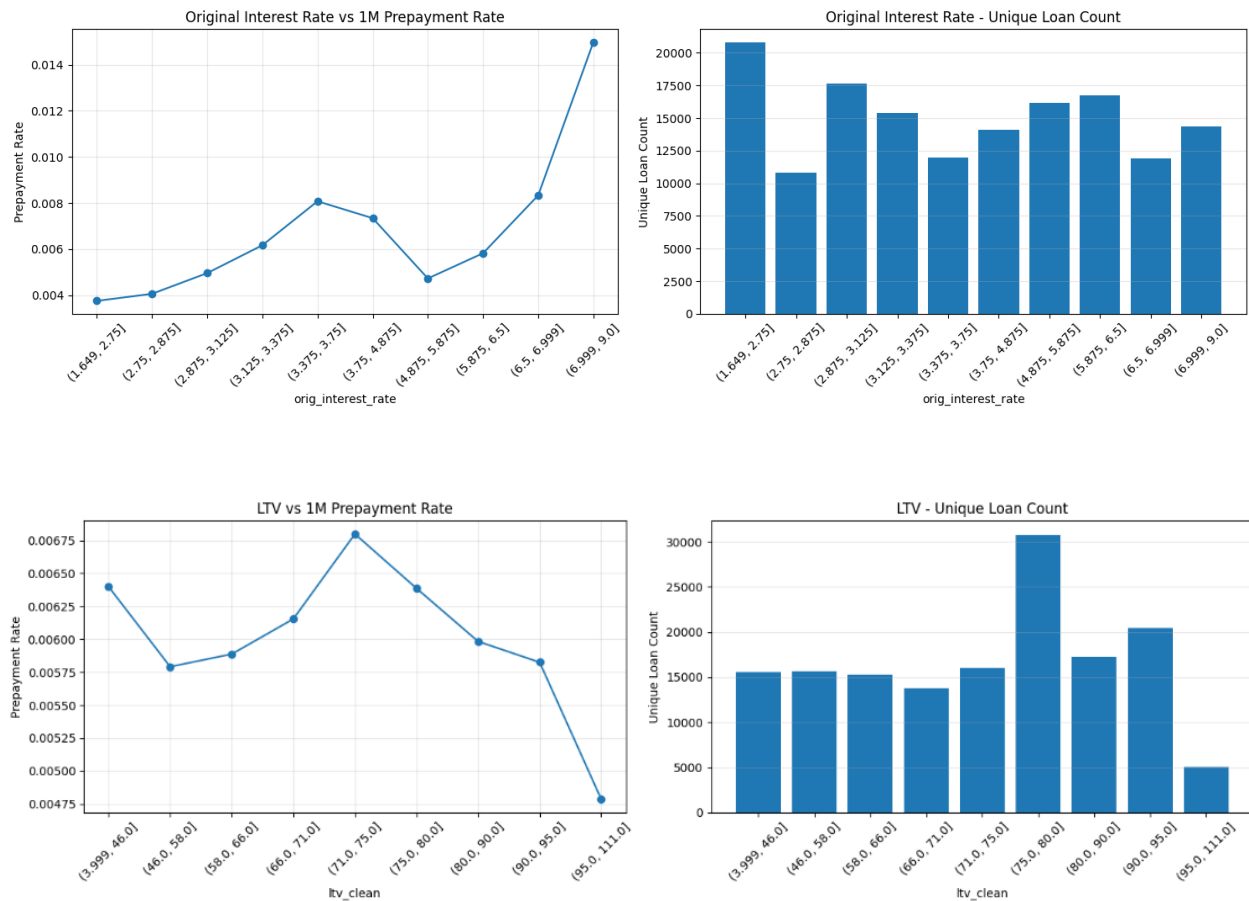
- Distinguished voluntary payoff from other zero-balance outcomes instead of labeling every mortgage termination as prepayment.
- Used forward-looking monthly records to construct 1-, 3- and 6-month targets and applied censoring so observations without sufficient future history were not labeled as non-events.
- The final feature panel contains 6,267,841 eligible 1-month observations and 38,374 positive next-month prepayment events.

Features

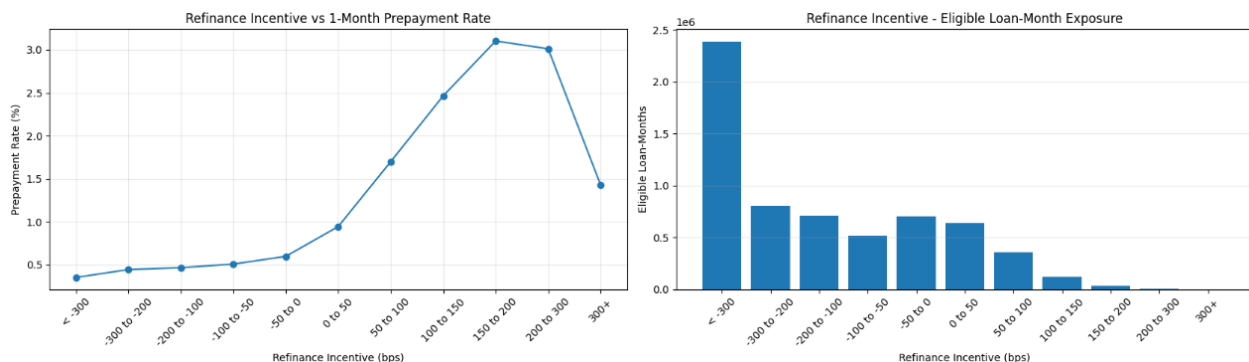
Feature family	Examples
Borrower / origination	FICO, DTI, LTV, Original UPB, Original interest rate, loan purpose, property type
Loan State	Current actual UPB, Current interest rate, loan age, months to maturity, current to actual UPB, original principal paydown pct, rate change since origination
Behavior	Recent and historical delinquency, modification history
Market	PMMS, refinance incentive, calendar month
Burnout	Prior exposure to favorable refinance windows, prior maximum incentive

2. Borrower behavior

Going to do Exploratory data Analysis (EDA) by static features and time varying features.



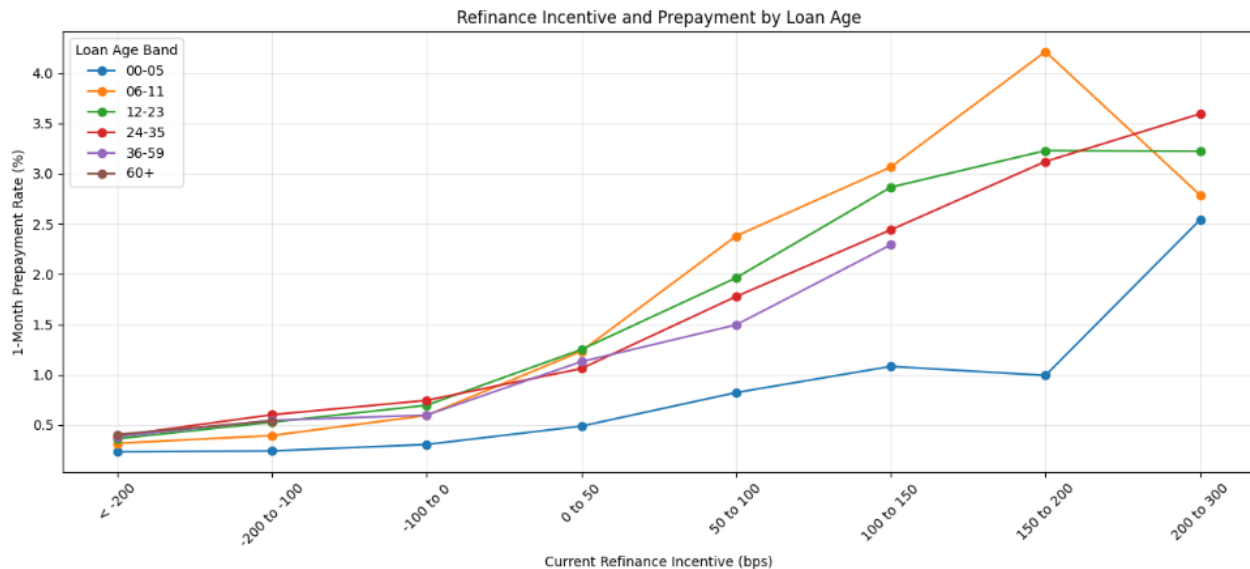
Prepayment varies materially across coupon and LTV segments, but the relationship is nonlinear.



- Loans less than 300 bps compared to PMMS prepaid at about 0.35% per month while the 100 to 150 bps above PMMS prepaid at about 2.46% about 7 times faster.
- The 150 to 200 bps band reached about 3.1% monthly prepayment although high-incentive observations become less common as the incentive increases.

- Refinance incentive is not the only driver. Loan seasoning, remaining balance, borrower characteristics, geography and prior performance history create meaningful heterogeneity **among borrowers facing similar rates**.

A borrower with a coupon materially above the prevailing mortgage rate has an economic reason to refinance. But transaction costs, borrower credit, remaining balance, loan age and prior refinance opportunities determine whether that incentive turns into an actual payoff.



0–5 month loans have substantially lower prepayment propensity across most of the incentive range. Even when refinancing becomes attractive, these extremely young mortgages react less strongly.

6–11 and 12–23 month loans appear particularly responsive to refinance incentive. Once incentive moves above roughly +50 to +100 bps, their prepayment rates accelerate substantially.

24–35 month loans also show a strong positive response.

For 36–59 month loans, the relationship appears positive through the ranges where there is enough data, but high-incentive observations become scarce.

And the 60+ group cannot really be used to evaluate positive refinance incentive in this sample because almost all of those observations occur in negative-incentive environments.

3. Modeling and Locked Out-of-Time Validation

Temporal design

Partition	Reporting period
Train	2020 to 2023
Validation	January to December 2024
Locked test	January 2025 to Feb 2026

Because this data is longitudinal, the same mortgage may appear in multiple calendar partitions as it ages. Loan ID is not a feature. This design evaluates future-period portfolio forecasting.

Model comparison on 2024 validation

Model	ROC-AUC	PR-AUC	Mean predicted	Actual
Logistic regression	0.638	0.0168	0.780%	0.543%
LightGBM Core	0.6997	0.0258	0.573%	0.543%
LightGBM + burnout	0.697	0.0269	0.488%	0.543%
XGBoost Core	0.700	0.0253	0.559%	0.543%

Initial benchmarking on the 2024 out-of-time validation period compared Logistic Regression, LightGBM Core, LightGBM with burnout features, and XGBoost Core.

Both gradient-boosting models materially outperformed Logistic Regression. Baseline LightGBM and XGBoost produced very similar discrimination, so the baseline comparison alone was not used to select the final model.

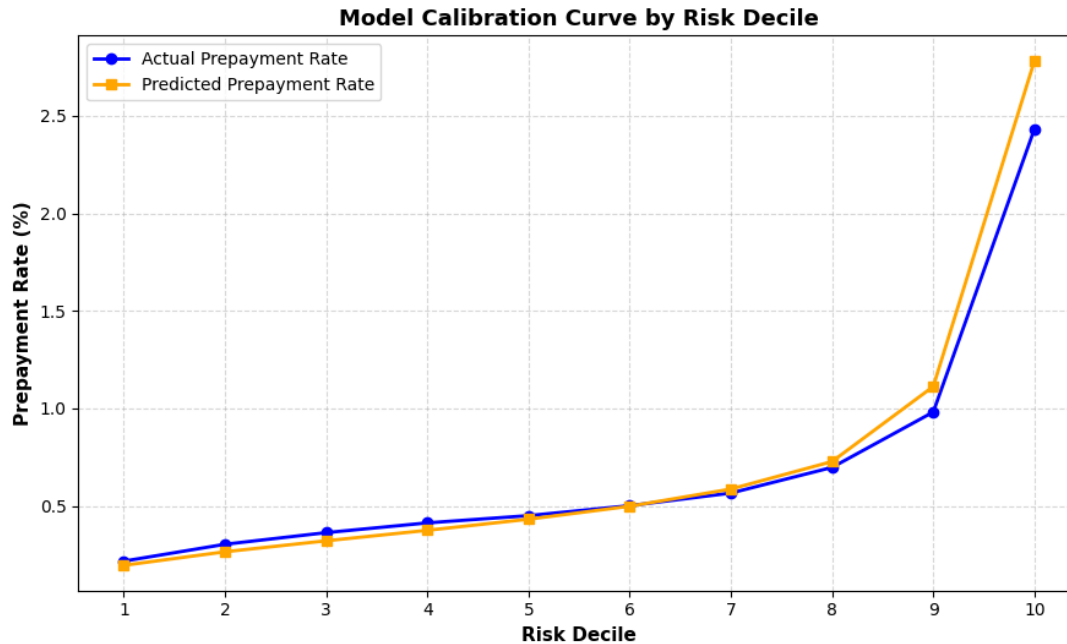
LightGBM was subsequently tuned on the 2024 validation period across alternative tree-complexity configurations.

The tuned LightGBM improved validation ROC-AUC from approximately 0.6997 to 0.7004, PR-AUC from 0.0258 to 0.0264, and log loss from 0.03178 to 0.03174. Its mean predicted prepayment rate of 0.562% was also closer to the realized validation rate of 0.543%.

This tuned LightGBM was then locked as the final model and evaluated on the untouched January 2025–February 2026 test period. No further tuning or feature selection was performed using the test set.

Locked test results

Test observations	Actual event rate	ROC-AUC	PR-AUC
1,642,132	0.693%	0.701	0.026



The model demonstrates strong risk ordering across the out-of-time test sample.

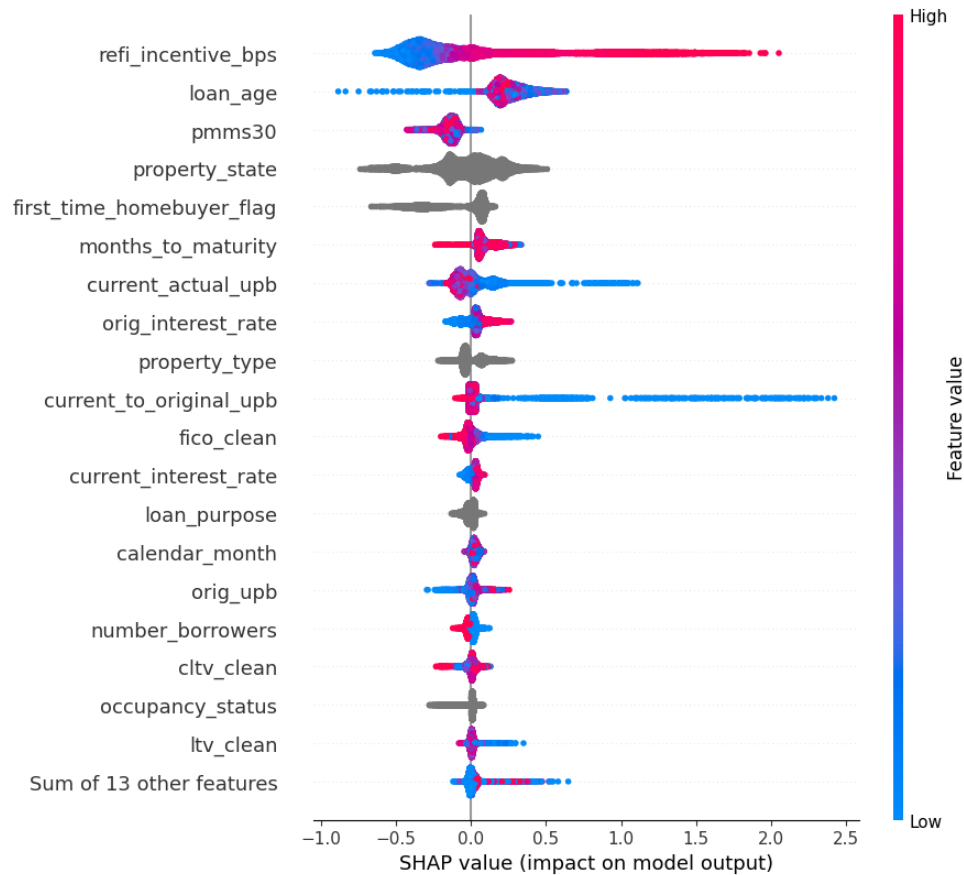
Realized one-month prepayment rates increase monotonically from approximately 0.22% in the lowest predicted-risk decile to 2.43% in the highest-risk decile.

The model slightly underpredicts realized prepayment rates in several lower-risk groups, while beginning to overpredict in the upper-risk tail.

The largest deviation occurs in the highest-risk decile, where the average predicted probability is approximately 2.78% compared with an actual prepayment rate of approximately 2.43%.

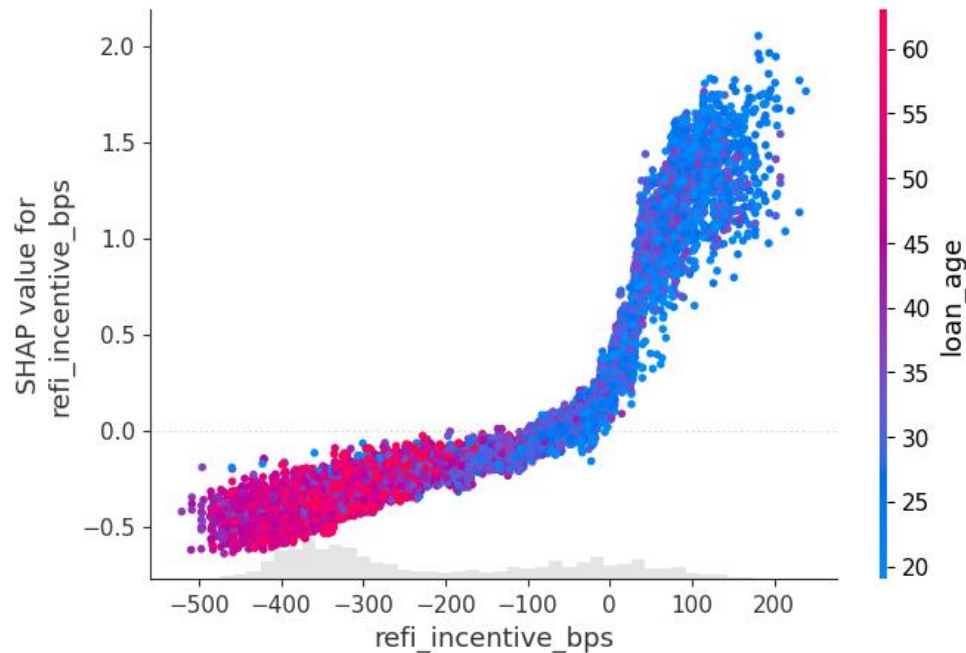
4. Explainability and the Loan-to-Pool Translation

To understand whether the model learned economically sensible relationships, I used SHAP to analyze the final tuned LightGBM model.



The SHAP beeswarm plot shows that refinance incentive is the most important driver of voluntary prepayment, followed by loan age, PMMS mortgage rate and several loan characteristics such as remaining maturity, current UPB and original interest rate.

This result is consistent with mortgage-finance intuition. Borrowers are more likely to refinance when their current mortgage coupon is high relative to prevailing market mortgage rates, making refinancing economically attractive.



For very negative incentives such as: -500 to -300 bps

There is mostly -

pink/purple → older loans, around 40–60+ months.

For positive incentives: 0 to +200 bps

There is mostly-

blue → younger loans, around 20–35 months

It reflects the historical structure of the mortgage population.

Older 2020–2021 mortgages originated when rates were extremely low. Younger 2022–2023 mortgages originated after rates rose.

This should not by itself be interpreted as a causal interaction between loan age and refinance incentive.

Loan to pool translation

For each active loan-month, the final model produced a predicted probability that the loan will voluntarily prepay in the following month.

$p_{i,t} = P(\text{voluntary prepayment in } t+1 \mid \text{information available at } t)$

To translate loan-level predictions into pool prepayment behavior, I aggregated the predicted probabilities using current unpaid principal balance (UPB) weights.

$$SMM_t = \frac{\sum_i UPB_{i,t} p_{i,t}}{\sum_i UPB_{i,t}}$$

This gives the expected single monthly mortality (SMM) for the pool in month t . Larger loans contribute more to pool cash-flow behavior than smaller loans, so the aggregation is balance-weighted rather than a simple average across loans.

This calculation was performed at the monthly reporting period level producing one predicted pool SMM for each month in the out-of-time test period.

The realized monthly SMM was computed using the corresponding realized voluntary prepayment outcomes.

To express prepayment in the more familiar annualized mortgage-market convention, the monthly SMM values were converted into CPR.

$$CPR_t = 1 - (1 - SMM_t)^{12}$$

This produced predicted and realized pool CPR time series across the holdout months.

Pool level forecast accuracy for both SMM and CPR was measured by –

- MAE
- RMSE
- Bias

CPR metrics

CPR MAE: 1.05 percentage points

CPR RMSE: 1.43 percentage points

CPR Bias: 0.01 percentage points

SMM metrics

SMM MAE: 0.095 percentage points

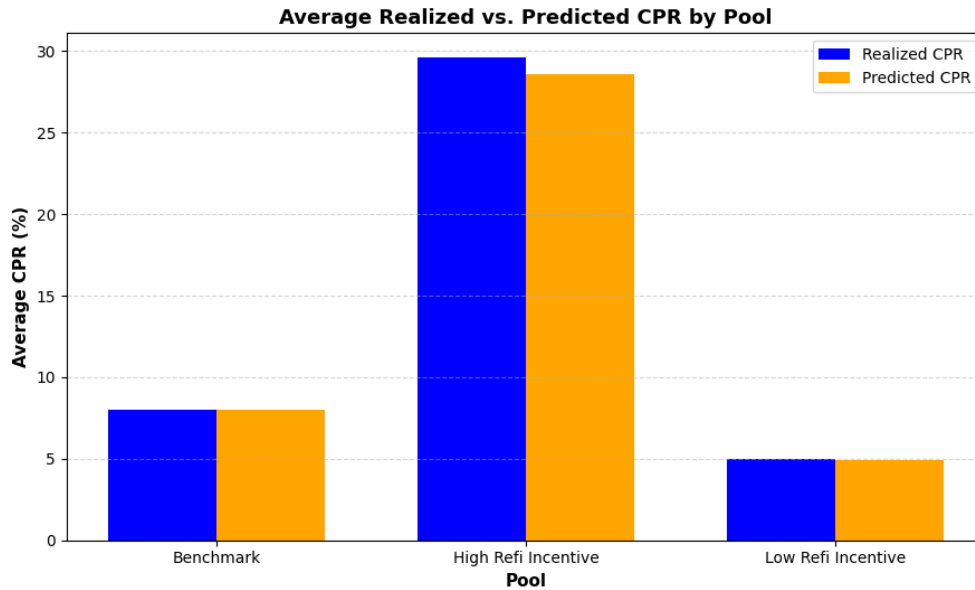
SMM RMSE: 0.130 percentage points

Pool-level validation: The ML model tracks realized prepayment behavior reasonably well over the holdout period, with a CPR MAE of 1.05 percentage points and essentially zero average bias.

Refi incentive Pool

To demonstrate economic usefulness, I created three illustrative mortgage pools based on refinance incentive:

- **Benchmark pool:** representative test population
- **High refinance incentive pool:** loans with strong refinance incentive
- **Low refinance incentive pool:** loans with little or no refinance incentive



The model correctly captures the sharp separation in prepayment behavior across these pools.

- Benchmark: realized ~8.0%, predicted ~8.0%
- High refi incentive: realized ~29.6%, predicted ~28.6%
- Low refi incentive: realized ~5.0%, predicted ~5.0%

The high-refinance-incentive pool prepays dramatically faster than the benchmark and low-incentive pools.

5. Cashflow Consequences: Prepayment changes when Principal Comes Back

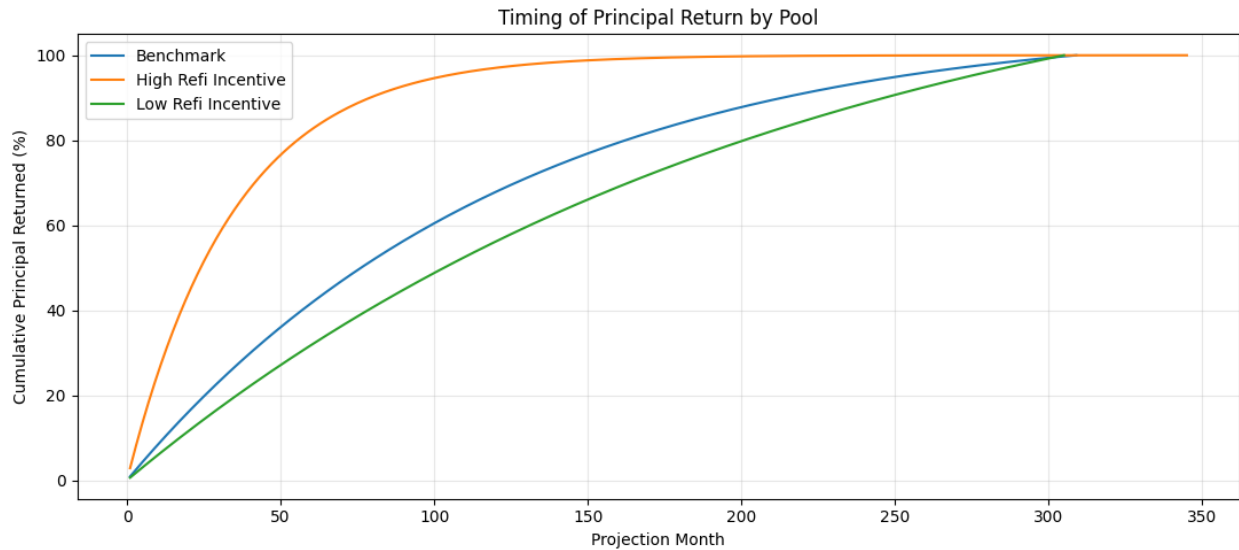
Business question: How do different prepayment speeds alter the timing of principal and interest received by an MBS investor?

Simplified deterministic pool cash-flow engine

- Used starting pool UPB, weighted average coupon (WAC), weighted average maturity (WAM) and ML derive pool SMM. Each pool's ML-derived SMM is held fixed through the deterministic forward cash-flow projection.
- Each month calculates scheduled payment, interest, scheduled principal, expected prepayment, total principal, ending UPB and total cash flow.
- Prepayment is applied to balance after scheduled principal: $\text{prepayment}_t = \text{SMM} \times (\text{beginning UPB} - \text{scheduled principal})$.

Cashflow result

Projection horizon	High refi	Benchmark	Low refi
12 months	29.3%	9.9%	7.1%
24 months	50.0%	19.0%	13.8%
60 months	82.5%	41.7%	31.9%
120 months	97.0%	67.8%	56.2%



For a given dollar of starting principal, prepayment primarily changes when principal is returned rather than whether it is ultimately returned. Faster prepayment reduces the balance available to generate future interest and creates contraction / reinvestment risk. Slower prepayment keeps principal outstanding longer and increases extension exposure.

Projected lifetime interest equals about 23.6% of starting UPB for the high-refi pool, 37.3% for the benchmark and 42.4% for the low-refi pool under the simplified constant-SMM projection.

6. Limitations

- Equal-quarter sampling rather than population weighting.
- Longitudinal temporal split can contain the same loan in multiple calendar partitions.
- Fixed SMM in forward projection.
- No stochastic rate paths/OAS
- Weaker discrimination in 2020-2021 vintages
- Overprediction in late 2025/early 2026
- Underprediction in some months like September 2025